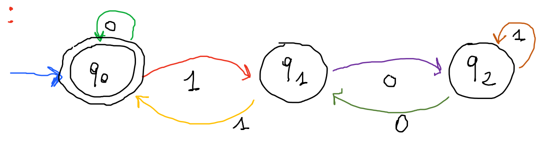
HOMEWORK

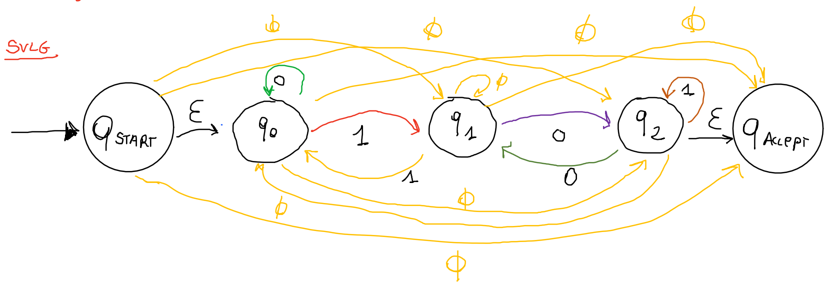
venerdì 18 marzo 2022 15:07

DATO IL DFA:

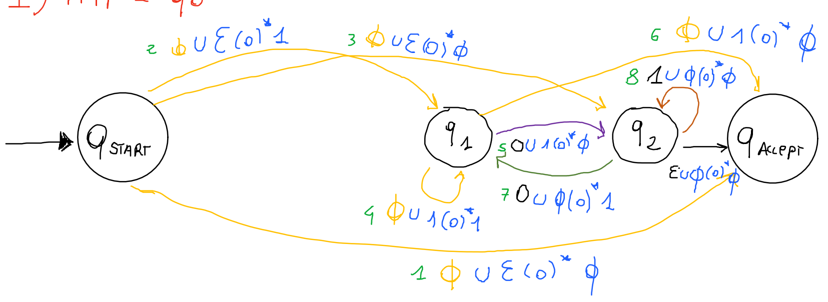


• Trasformiamo il dfa in una Rex!

SVLG



1) RIP = q_0



1 • Q_{START} raggiunge Q_{ACCEPT}, tramite q_0. $Q_{START} \to Q_{ACCEPT}$, tramite q_0.

2 • $Q_{START} \to Q_1$, tramite q_0.

3 • $Q_{START} \to q_2$, tramite q_0.

4 • $q_1 \to q_1$, tramite q_0.

5 • $q_1 \to q_2$, tramite q_0.

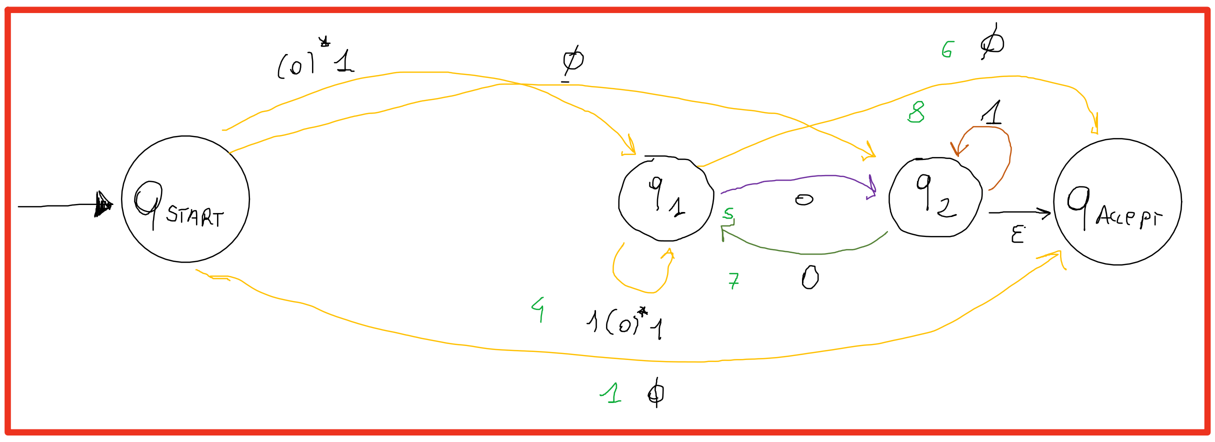
6 • $q_1 \to q_{ACCEPT}$, tramite q_0.

7 • $q_2 \to q_1$, tramite q_0.

8 • $q_2 \to q_2$, tramite q_0.

9 • $q_2 \to q_{ACCEPT}$, tramite q_0.

Riscriviamo il grafo!

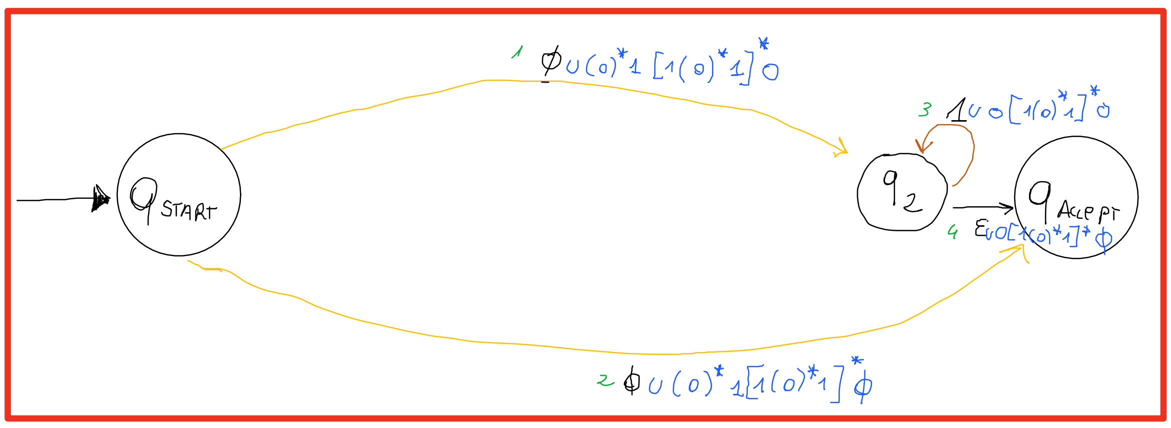


• $\varepsilon \cup \phi = \varepsilon$

• $b(a \cup b)^* \varepsilon = b(a \cup b)^*$

• $b(a \cup b)^* \phi = \phi$

2) RIP = q_1



1. $Q_{START} \to q_2$, tramite q_1.

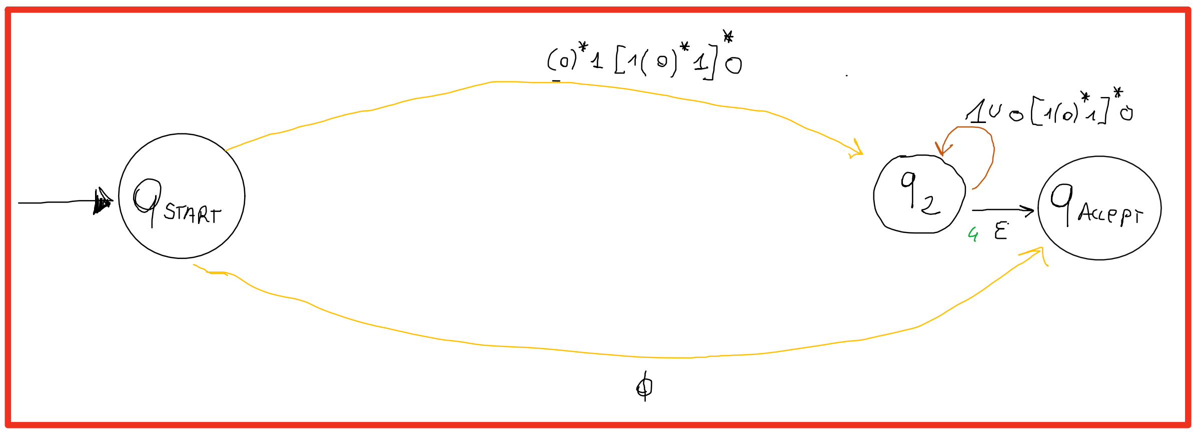
2. $Q_{START} \to q_{ACCEPT}$, tramite q_1.

3 $q_2 \to q_2$, tramite q_1.

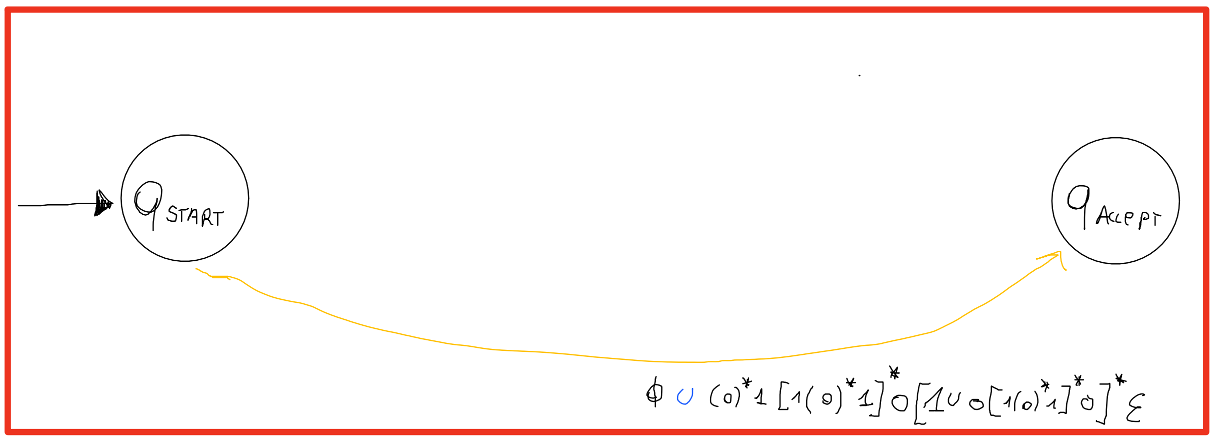
4. $q_2 \to q_{ACCEPT}$, tramite q_1.

Riscriviamo il grafo:

Risolviamo il grafo.



2) RIP = q_2



1 ∘ q start → q accept, tramite q_2!

$$\boxed{(0)^*1\,[1(0)^*1]^*\,0\,[1 \cup 0[1(0)^*1]^*0]^*\,\varepsilon}$$